Day – 3 NAT and PAT Concepts Theory.

Task – 1 Config the static NAT on router.

- Step 1: Setting up the Network Topology in Packet Tracer
- To simulate a basic Static NAT environment, build the following layout:
- Internal Network: Place PC 0 (Internal PC) and connect it via a switch to GigabitEthernet 0/0 on the router.
- External Network: Place PC 1 (External PC acting as an external server or remote host) and connect it directly or via a switch to GigabitEthernet 0/1 on the router.
- Step 2: Defining the IP Addressing Scheme
- Internal LAN (Private): 192.168.1.0/24
- PC 0 (Internal PC): 192.168.1.100 (Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1)
- Router Inside Interface (GigabitEthernet 0/0): 192.168.1.1 (Subnet Mask: 255.255.255.0)
- External WAN (Public): 203.0.113.0/24
- Router Outside Interface (GigabitEthernet 0/1): 203.0.113.1 (Subnet Mask: 255.255.255.0)
- PC 1 (External PC): 203.0.113.50 (Subnet Mask: 255.255.255.0, Default Gateway: 203.0.113.1)
- Mapped Public IP for Static NAT: 203.0.113.10 (This is the public face assigned to the internal PC during translation).
- Step 3: Configuring the End Devices (PCs)
- Configure PC 0 (Internal):
- Click PC 0 -> Desktop tab -> IP Configuration.
- Set IPv4 Address to 192.168.1.100, Subnet Mask to 255.255.255.0, and Default Gateway to 192.168.1.1.
- Configure PC 1 (External):
- Click PC 1 -> Desktop tab -> IP Configuration.
- Set IPv4 Address to 203.0.113.50, Subnet Mask to 255.255.255.0, and Default Gateway to 203.0.113.1.
- Step 4: Configuring the Router Interfaces (Inside and Outside)
- Click on the router, open the CLI tab, and assign the IP addresses while designating which interface faces the private network (nat inside) and which faces the public network (nat outside):
- Router> enable
- Router# configure terminal

- Router(config)# hostname RouterNAT
- ! Configure the Internal (Inside) Interface
- RouterNAT(config)# interface gigabitEthernet 0/0
- RouterNAT(config-if)# ip address 192.168.1.1 255.255.255.0
- RouterNAT(config-if)# ip nat inside
- RouterNAT(config-if)# no shutdown
- RouterNAT(config-if)# exit
- ! Configure the External (Outside) Interface
- RouterNAT(config)# interface gigabitEthernet 0/1
- RouterNAT(config-if)# ip address 203.0.113.1 255.255.255.0
- RouterNAT(config-if)# ip nat outside
- RouterNAT(config-if)# no shutdown
- RouterNAT(config-if)# exit
- Step 5: Configuring Static NAT Translation
- Set up the static translation rule mapping the internal private IP (192.168.1.100) to the designated public IP (203.0.113.10):
- RouterNAT(config)# ip nat inside source static 192.168.1.100 203.0.113.10
- RouterNAT(config)# exit
- RouterNAT# write memory

Task – 2 Config Port Address Translation(PAT).

- Step 1: Setting up the Network Topology in Packet Tracer
- To simulate a Port Address Translation (PAT) or NAT Overload environment, build the following layout:
- Internal Network: Place PC 0 and PC 1 (Internal PCs), connect them to a switch, and link that switch to GigabitEthernet 0/0 on the router.
- External Network: Place a server or another PC acting as an external Web Server, and connect it to GigabitEthernet 0/1 on the router.
- Step 2: Defining the IP Addressing Scheme
- Internal LAN (Private): 192.168.1.0/24
- PC 0: 192.168.1.10 (Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1)
- PC 1: 192.168.1.20 (Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1)
- Router Inside Interface (GigabitEthernet 0/0): 192.168.1.1 (Subnet Mask: 255.255.255.0)
- External WAN (Public): 203.0.113.0/24
- Router Outside Interface (GigabitEthernet 0/1): 203.0.113.1 (Subnet Mask: 255.255.255.0)

- External Web Server: 203.0.113.100 (Subnet Mask: 255.255.255.0, Default Gateway: 203.0.113.1)
- Step 3: Configuring the Router Interfaces
- Click on the router, go to the CLI tab, and configure the internal and external interfaces, marking them with the correct NAT directions:
- Router> enable
- Router# configure terminal
- Router(config)# hostname RouterPAT
- ! Configure the Internal (Inside) Interface
- RouterPAT(config)# interface gigabitEthernet 0/0
- RouterPAT(config-if)# ip address 192.168.1.1 255.255.255.0
- RouterPAT(config-if)# ip nat inside
- RouterPAT(config-if)# no shutdown
- RouterPAT(config-if)# exit
- ! Configure the External (Outside) Interface
- RouterPAT(config)# interface gigabitEthernet 0/1
- RouterPAT(config-if)# ip address 203.0.113.1 255.255.255.0
- RouterPAT(config-if)# ip nat outside
- RouterPAT(config-if)# no shutdown
- RouterPAT(config-if)# exit
- Step 4: Configuring an Access Control List (ACL)
- Create a standard ACL to identify which internal private IP addresses are permitted to undergo NAT translation:
- RouterPAT(config)# access-list 1 permit 192.168.1.0 0.0.0.255
- Step 5: Enabling PAT with the Overload Command
- Configure the router to translate traffic matching Access List 1 using the IP address of the external interface (GigabitEthernet 0/1), leveraging the overload keyword so multiple internal devices can share that single public IP using unique source ports:
- RouterPAT(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
- RouterPAT(config)# exit
- RouterPAT# write memory

Task – 3 Test your NAT config.

- To verify your PAT configuration, follow these steps to generate traffic and inspect the translation table on the router:
- 1. Send a Ping from the Internal PC

- Open PC 0 (IP 192.168.1.10) or PC 1 (IP 192.168.1.20), go to the Desktop > Command Prompt, and ping the external Web Server (203.0.113.100):
- C:\> ping 203.0.113.100
- Ensure you receive successful echo replies. This live traffic triggers the router to create dynamic NAT translation entries.
- 2. Capture and View the NAT Translation Table on the Router
- Click on your router, go to the CLI tab, and enter the following command to display active translations:
- RouterPAT# show ip nat translations
- What to Look For in the Output:
- You will see a table showing how the router translates traffic. Because you are using Port Address Translation (PAT / Overload), the output will look similar to this:
- Pro Inside global Inside local Outside local Outside global
- icmp 203.0.113.1:1 192.168.1.10:1 203.0.113.100:1 203.0.113.100:1
- icmp 203.0.113.1:2 192.168.1.20:1 203.0.113.100:2 203.0.113.100:1
- Inside local: The private IP address of your internal PC (192.168.1.10 or 192.168.1.20) before translation.
- Inside global: The single public IP address of the router's outside interface (203.0.113.1) paired with a unique port number (e.g., :1, :2) assigned to distinguish between the different internal devices sharing the same public IP.
- Outside global / local: The public IP address of the external Web Server (203.0.113.100) that received the ping.

Task – 4 Modify your packet tracer topology.

- Step 1: Updating the Network Topology in Packet Tracer
- To add a third internal PC to your existing PAT setup:
- Connect PC 2 (the third internal PC) to the existing switch on the internal network using a copper straight-through cable.
- Ensure the router's internal interface (GigabitEthernet 0/0) and the external interface (GigabitEthernet 0/1) remain connected as they were configured previously.
- Step 2: Configuring the Third Internal PC
- Assign a private IP address within the same local subnet (192.168.1.0/24) to PC 2:
- Click PC 2 -> Desktop tab -> IP Configuration.

- IPv4 Address: 192.168.1.30
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1 (matching the router's inside interface).
- Step 3: Verifying and Updating the Access Control List (ACL)
- Since your existing Access List (Access-List 1) was set to permit the entire subnet 192.168.1.0 0.0.0.255, PC 2 is already automatically covered under the rule. No modifications to the ACL or the ip nat inside source list … overload command are required because the wildcard mask encompasses all hosts from 192.168.1.0 through 192.168.1.255.
- Step 4: Testing Connectivity from All Three PCs
- To generate traffic and populate the NAT translation table, open the command prompt on each of the three internal PCs and ping the external Web Server (203.0.113.100):
- From PC 0 (192.168.1.10):
- C:\> ping 203.0.113.100
- From PC 1 (192.168.1.20):
- C:\> ping 203.0.113.100
- From PC 2 (192.168.1.30):
- C:\> ping 203.0.113.100
- Ensure you receive successful echo replies from all three devices.
- Step 5: Verifying the Updated NAT Translation Table
- Click on your router, go to the CLI tab, and view the active NAT translation table to confirm that all three internal PCs are sharing the single public IP using Port Address Translation:
- RouterPAT# show ip nat translations
- Expected Output Structure:
- You will see unique translation entries for each internal source IP mapped to the router's single public interface IP (203.0.113.1) via individual source ports:
- Pro Inside global Inside local Outside local Outside global
- icmp 203.0.113.1:1 192.168.1.10:1 203.0.113.100:1 203.0.113.100:1
- icmp 203.0.113.1:2 192.168.1.20:1 203.0.113.100:2 203.0.113.100:1
- icmp 203.0.113.1:3 192.168.1.30:1 203.0.113.100:3 203.0.113.1

Task – 5 Explain how NAT and PAT are limited in public ip addresses.

- How NAT and PAT Solve the IPv4 Public Address Shortage

- There are far more active internet-connected devices in the world—smartphones, laptops, smart TVs, and IoT gadgets—than there are available IPv4 public addresses. If every single smartphone running the Flipkart or Zomato app required its own unique public IP address to browse products or order food, the global internet would have run out of addresses decades ago.
- NAT (Network Address Translation) and PAT (Port Address Translation) solve this problem by allowing an entire household, office building, or mobile carrier tower to share a single public IP address.
- NAT maps an entire pool or individual private IP addresses to public ones, but traditional NAT usually requires a 1-to-1 ratio for public-to-private IPs, which doesn't fully solve the shortage.
- PAT (or NAT Overload) takes this a step further. It allows thousands of devices behind a local router to share just one public IP address simultaneously. When your phone sends a request to Flipkart's servers, the local router rewrites the source private IP address to its own public IP address and appends a unique source port number to the packet. When Flipkart's server replies, it sends the data back to that public IP and port combination. The router looks at its translation table, remembers which internal device originally made the request, and forwards the data back to your phone.
- Without PAT, every single user browsing Zomato from home or mobile data would need their own dedicated public IP address, making modern internet scale completely impossible.
- Real-World Scenarios Where Incorrect NAT/PAT Setup Breaks Connectivity
- Overloaded Translation Tables or Port Exhaustion during Flash Sales (e.g., Flipkart Big Billion Days)
- The Scenario: During a massive online sale event, thousands of users behind a corporate network or carrier-grade NAT (CGNAT) gateway are simultaneously hitting Flipkart's servers.
- The Problem: If the gateway router's NAT configuration lacks proper port allocation or has an overly restrictive translation timeout, the router can run out of available unique source ports to assign to incoming connection requests (port exhaustion). As a result, new connection attempts fail, users get stuck on loading screens, and carts cannot load during checkout because the router simply cannot translate any more simultaneous outbound connections.
- Missing or Misconfigured Outbound Overload Rules / ACLs
- The Scenario: A new branch office network is set up for employees, and they try to use web apps like Zomato to order lunch during office hours, but none of the internal PCs can load any external web pages.
- The Problem: If the network administrator configures the router's inside and outside interfaces correctly but forgets to include the overload keyword or misconfigures the Access Control List (ACL) matching the private subnet, the

router drops or fails to translate outgoing traffic. Packets leaving the internal PCs carry private, unroutable IP addresses (192.168.x.x) out onto the public internet. Because public routers drop private IP traffic instantly, the connection fails entirely, resulting in complete internet isolation for those users.